Intelligent Reasoning Systems



Team OptEV (Group 25)

- Ghanashyam Ramia Kasthurirangan Pandurengan (A0357701E)
- Gurneet Kaur Bhuller (A0354164E)
- Shrey Yadav (A0354457W)
- Sushant Maramkulam Shiji (A0354938M)



About OptEV+

A dynamic reasoning system that supports the planning of EV charging infrastructure across Singapore, drawing on charging demand, existing coverage, car park capacity, accessibility and site feasibility.

Market Situation

Singapore's **2030 EV** goal

Growing **EV Demand**

Expanding **Charger Infrastructure**



Project Background

Turning live data into charging decisions

Our project aims to develop a Dynamic Intelligent EV Charging Recommendation System to help Singapore's government identify and rank the most suitable locations for new EV charging infrastructure.

Process automation

Scores candidate sites (HDB and public car parks, hawker centres, petrol stations and existing charging sites) on charging pressure, coverage, parking capacity and site constraints. It recommends where to add charging points, how many and what type, and logs a reason for every site accepted, rejected or deferred.

Data integration

Combines live charger and car park feeds with the car park registry and planning-area boundaries into one time-series baseline. Matching chargers to car parks by location gives utilisation, saturation and charging pressure. Scenarios then project this forward along Singapore's published deployment targets.

Performance monitoring

When a planner selects a site, it is saved as planned capacity, kept separate from observed data, and the scenario re-solves so the next recommendation accounts for it. Nothing observed is overwritten, so every choice stays reversible. Growth beyond the collection window comes from simulated future scenarios.

Market Context

01
Background

Singapore targets 60,000 charging points by 2030, 40,000 of them in public car parks. About 34,700 were in place by September 2026, serving more than 82,100 EVs.

02
Problem

Over 90% of HDB car parks already have chargers. Next, planners must choose which sites get more, how many points each needs, and how to spend a limited budget.

03
Market Research

Research models usually pick sites once, from static data. Few decide how many chargers each site needs, and driver apps only show where to charge right now.



OptEV+ closes this gap: it turns live government data into a ranked, budget-aware plan with a reason for every site, and updates as planners make choices.

Market Research

01 · Data Gap
Live feeds, no unified pressure view
→ Area-level pressure indicators

02 · Planning Gap
Utilisation ≠ optimal allocation
→ Constrained optimisation

03 · Explainability Gap
Decision reasons aren't retained
→ Traceable decision records

Our Market Research

EXISTING SOLUTIONS

- PlugShare
- SG EV 2030
- Charge+
- More Chargers
- EV Growth
- SP Mobility

Market Gap

No intelligent government planning platform
for dynamic EV charger deployment

Our Solution

Dynamic AI Reasoning + Optimization +
Explainable EV Infrastructure Planning

Island-wide, 1 Area in Depth

All areas covered. For simulation purposes, 1 area selected

Public Charging Only

Publically accessible locations only

Today Measured, 2030 Simulated

Live baseline, simulated future

Project Scope & Deliverables

Planning Web Application

Map → Rank → Budget
Allocate → Select

Explainable Answers

“Why this site?” with evidence

Report, Code and Videos

Evaluation
Code Guide
2 Videos

Data Collection

01

Static site records

Baseline data that rarely changes: HDB car park locations and lot capacity, land-use zoning, planning-area boundaries and nearby amenities. It tells the system what each site can physically hold and what surrounds it.

02

Live feeds, every 15 minutes

A Python collector calls the LTA DataMall EV charging API and the data.gov.sg HDB car park availability API every 15 minutes. Each timestamped snapshot captures charger status and free lots island-wide, with the raw data saved as a backup.

03

Matching and quality checks

Chargers and car parks are linked by location and car park number, and every site is assigned to one of the 55 planning areas. Each run reports coverage (how many sites reported) and freshness (how recent the data is), so gaps are flagged, not hidden.

System Design

Five stages, as in the proposal: government data in; a ranked, constrained and explained allocation out.

1 Data foundation

Collect, validate, link

Foundation

- **Charging point status** — LTA DataMall · every 15 min
- **Car park availability** — LTA + HDB · every minute
- **Car park registry** — HDB info · URA capacity
- **Boundaries & land use** — Master Plan 2025
- **Candidate sites** — hawker · CC · petrol · malls
- **Context & growth** — residents · registrations

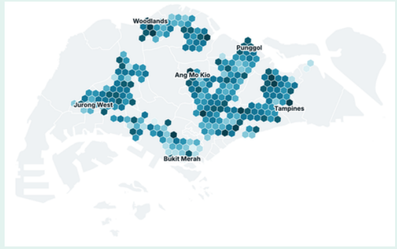
Align coordinates · match at 50/100/200 m · assign planning area
→ **one candidate-site table**

Python · PostGIS · pyproj

2 Measurement & inference

Measure use, infer demand

Knowledge discovery



Real data: HDB car park use at 3 am

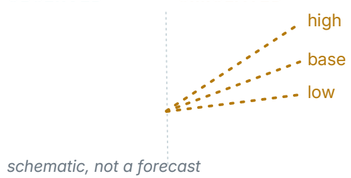
- Utilisation and saturation per site, per hour
- Demand inferred from occupancy, with a stated reliability bound
- Estimates carried to sites with no chargers

pandas · SciPy · scikit-learn

3 Scenario generation

As per 2030 Requirement

Simulated layer



schematic, not a forecast

- Demand projected forward under declared growth assumptions
- Synthetic records generated at real candidate locations
- Simulated data labelled, never merged with observed

NumPy · seeded scenarios

4 Reasoning & optimisation

Rules first, then allocate

Decision automation · Optimisation

RULES

Accept | Conditional | Defer | Reject

each with a reason code

ALLOCATION · 6 NEW POINTS



Punggol Way has no chargers today

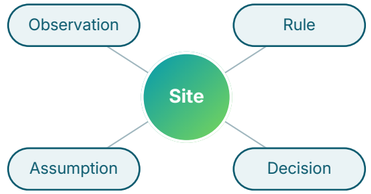
Independent check of every constraint before release

Rule engine · OR-Tools CP-SAT

5 Explanation & interface

Show the answer, and why

Cognitive techniques



"Why Punggol Way?"
Cites rule, data and assumptions

- Linked evidence behind every decision
- Ranked map view with selection controls
- GraphRAG will be done for natural language answers.

Neo4j · LangGraph · Streamlit

Planner selects a site → recorded as planned capacity → the scenario re-solves

THREE DATA LAYERS, NEVER MERGED

- **OBSERVED** measured from the feeds
- **SIMULATED** generated under assumptions
- **PLANNED** chosen by the planner

Outcome & Evaluation

Every stage of the build is tested on its own. Metrics and targets are fixed now, before anything is built.

1

KNOWLEDGE DISCOVERY

Demand Inference Validation

Q: Can we recover demand from occupancy, and rank it where no charger exists?

METRICS

MAE = 1/n Σ|y − ŷ|, utilisation on hidden days

MAPE = 1/n Σ|y − ŷ| ÷ y, vs simulated demand

Spearman ρ = 1 − 6Σd² ÷ n(n² − 1), site ranks

DATA SPLIT

Hidden days (temporal); hidden sites and areas (spatial)

EXAMPLE

Bedok North, 85% busy → 2.65 drivers/h, 1.52 turned away. Above ~93% busy: a lower bound

BASELINE · TARGET

Same hour yesterday; area average. Beat both; ρ ≥ 0.5

2

GENERATED DATA

Scenario Validation

Q: Does the plan still hold if our growth assumption is wrong?

METRICS

Top-10 stability = sites still in top 10 ÷ 10

Shifts tested growth ±20%, cost ±20%, demand ±10%

Back-cast MAPE on recent months hidden while fitting

ANSWER KEY

None: it measures sensitivity, not truth

HOW TO READ IT

8 of 10 stay → stable. 5 of 10 → the plan hinges on the assumption

TARGET

Flag any shift keeping fewer than 7

3

DECISION AUTOMATION + OPTIMISATION

Rules & Allocation Validation

Q: Do rules decide like a planner, and does the optimiser beat simpler methods?

METRICS

Rule accuracy = correct ÷ 30 test sites set in advance (κ ≥ 0.7)

Violations = 0; checker catches every seeded fault

Uplift = (D_ours − D_base) ÷ D_base, D = demand covered

BASELINES

A busiest first · B population share · C random ×100

EXAMPLE · 5 SITES, 6 POINTS

Ours rescues 1.71 drivers/h vs A 1.55 (+10%), C 1.35 (+27%)

TARGET

100% rules; 0 violations; beat A, B, C

4

COGNITIVE TECHNIQUES

Explanation Validation

Q: Are answers correct, and is every claim backed by evidence?

METRICS

Answer accuracy = correct ÷ 20 questions

Citation rate = answers citing evidence ÷ answers

Unsupported claims = numbers absent from the evidence

Precision@k = relevant in top k ÷ k

ANSWER KEY

Written in advance from source tables, not the graph

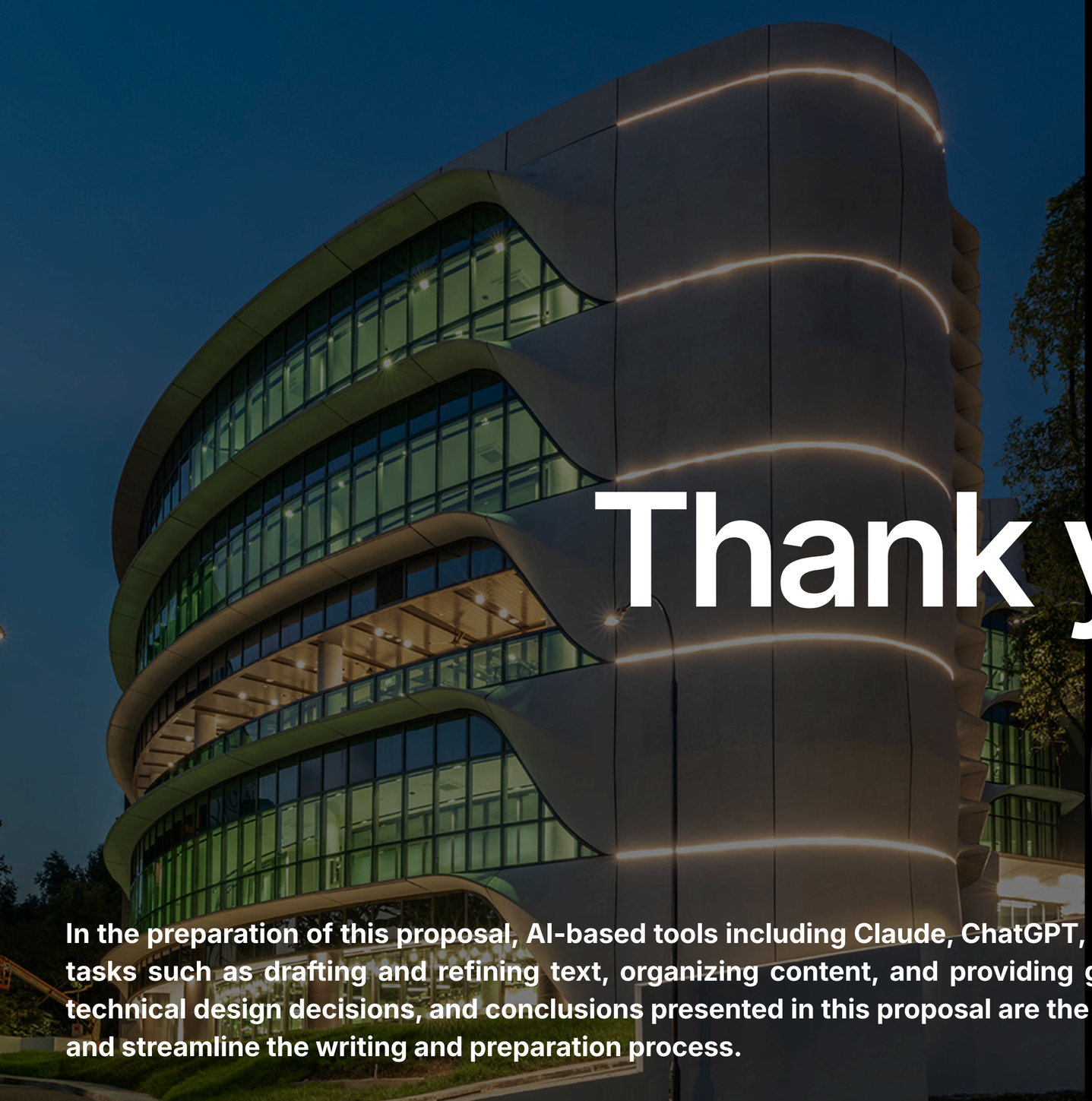
EXAMPLE

"Why Punggol Way?" must cite its rule and observations

TARGET

≥ 90% correct; 100% cite; 0 unsupported

End to end, one number: unmet demand covered per new charging point, against the best baseline under the same budget.



Thank you!

In the preparation of this proposal, AI-based tools including Claude, ChatGPT, and Gemini were used in a supporting capacity to assist with tasks such as drafting and refining text, organizing content, and providing general research and reference support. All ideas, analysis, technical design decisions, and conclusions presented in this proposal are the work of the project team, with AI tools used solely to support and streamline the writing and preparation process.